

Customer 360

From raw transactions to production-ready customer intelligence: an end-to-end analysis of the Online Retail II dataset.

Online Retail II (UCI) · 1,033,036 clean transactions · Dec 2009 – Dec 2011 · All values GBP

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1. Executive summary

This project takes two years of transaction logs from a UK-based online gift retailer and builds the full analytical stack on top of them: a cleaned and documented data foundation, a queryable SQL layer, exploratory and segmentation analysis, a predictive model for repeat purchase, four competing recommendation engines, a market basket analysis, and the engineering scaffolding needed to run all of it again.

1.03M

Clean transactions

£18.9M

Total revenue

5,942

Identified customers

5,304

Distinct products

Four results carry most of the practical weight.

Revenue is concentrated and seasonal. The United Kingdom accounts for 84.8% of revenue. Both years in the sample peak in November, and the September-to-November window is where the year is decided. A UK-specific shock has nothing to offset it.

Customers split cleanly into three behavioural groups. K-Means on log-scaled RFM features selects three segments. An active core of 2,316 customers — 39% of the base — spends roughly nine times what the 3,521 lapsing customers do. A third group of 105 customers

is net-negative: their returns exceed their purchases.

Repeat purchase is predictable, and recency is the signal. A Gradient Boosting model trained on eight behavioural features reaches 0.772 PR-AUC and 0.800 ROC-AUC on a held-out test set. Roughly 60% of the model's decision comes from a single feature: how long ago the customer last ordered.

Purchase order beats purchase similarity for recommendations. A sequence model that reads what a customer bought most recently achieves 29.8% Hit Rate@10, against 10.2% for classical matrix factorisation on the same held-out targets — a 2.9× improvement, and the single largest performance gap in the project.

HOW TO READ THIS REPORT

Every figure quoted here is produced by the analysis notebook rather than estimated, and each section states the protocol that generated it. Where a result is fragile or easily over-read — the truncated final month, the inflated lift values in the basket analysis, the definition of a "basket" — the report says so at that point rather than in a footnote.

2. Data foundation and preparation

The raw extract holds 1,067,371 rows across eight columns, spanning 1 December 2009 to 9 December 2011. Before any of it could support a model, it needed real repair.

Cleaning decisions

Duplicates. 34,335 exact duplicate rows were removed, leaving 1,033,036. This happens first, before any aggregation, so that duplicated line items cannot inflate customer revenue or product demand.

Missing customer identifiers. 22.8% of rows — 235,151 of them — carry no customer ID. These are guest or untracked checkouts, a known characteristic of this dataset. Dropping them would understate total revenue by roughly a fifth, so they are retained for financial aggregates and excluded only from customer-level modelling. Every customer-level figure in this report is therefore computed on the 5,942 identified customers, while revenue totals cover all transactions.

Returns and cancellations. Negative quantities are flagged as returns (22,496 transactions) and invoices with a **c** prefix as cancellations (19,104). Both are isolated rather than silently dropped, which allows the net-negative customer segment to surface later instead of being cleaned away.

Typing and derived fields. Column names are standardised, dates and prices are coerced to proper types, and revenue is computed as quantity multiplied by unit price. Date parts — year, month, day, weekday, hour — are derived once and reused throughout. The final transaction table carries 18 typed columns, each documented in a generated data dictionary.

Table 1 — Outputs of the preparation stage

TABLE	ROWS	GRAIN AND CONTENTS

transactions	1,033,036	Cleaned transaction line items
customers	5,942	Orders, revenue, lifetime, returns, product breadth
products	5,304	Demand, customer reach, return counts
daily_sales	604	Daily revenue, quantity, invoices, customers

Each table is written to CSV and reloaded into the database, so the entire analysis is rebuildable from the raw archive without manual intervention.

3. The analytics layer

The processed outputs are loaded into an indexed SQLite database rather than left as loose files. Indexes cover `customer_id`, `stock_code`, `invoice_date_only` and both revenue columns, so the analytical joins stay fast as the transaction table grows.

Ten analytical queries exercise the patterns the downstream work depends on: aggregation and filtering, JOINS across customer and product grain, common table expressions, and window functions. The month-over-month revenue query is representative:

```
WITH monthly_revenue AS (
    SELECT invoice_year, invoice_month,
           SUM(revenue) AS revenue
    FROM transactions
    GROUP BY invoice_year, invoice_month
)
SELECT invoice_year, invoice_month, revenue,
       LAG(revenue) OVER (
           ORDER BY invoice_year, invoice_month
       ) AS previous_month_revenue
FROM monthly_revenue
ORDER BY invoice_year, invoice_month;
```

Others rank customers by revenue within their country using `RANK() OVER (PARTITION BY ...)`, compute a seven-day rolling revenue window, and derive product return rates with a null-safe denominator. The layer exists so that commercial questions can be answered directly in SQL without re-running the Python pipeline.

4. Commercial picture

Revenue is sharply seasonal

Monthly revenue moves in a strong annual cycle. November is the peak in both complete years — £1.42M in 2010 and £1.46M in 2011 — roughly 2.4 times the quietest month in the series. The build-up runs from September, with October and November together carrying a disproportionate share of the year.

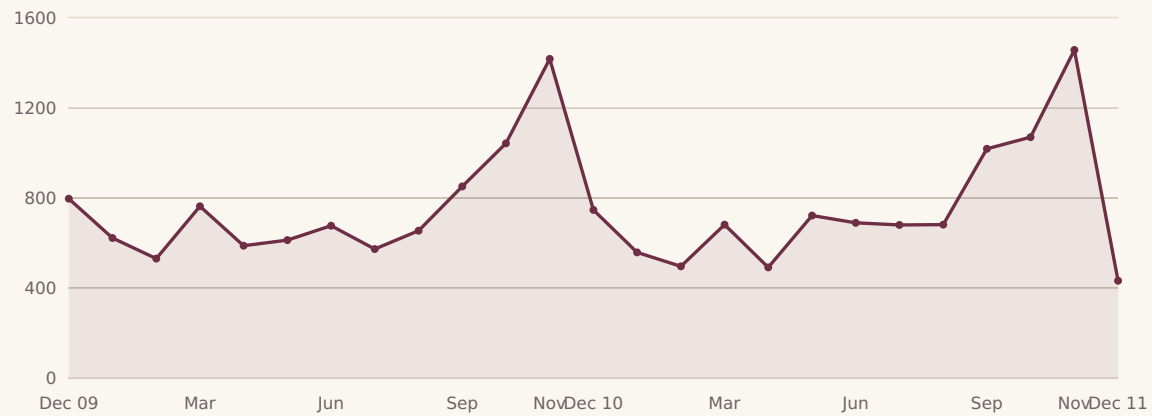


Figure 1 — Monthly revenue, December 2009 to December 2011 (£ thousands). The final data point is partial: the source data ends on 9 December 2011, so the apparent collapse in that month is an artefact of the extract, not a commercial event.

The operational reading is that stock, staffing and campaign spend decisions made in late summer determine the year's outcome. The largest month-over-month gain in the series is the November 2011 step of £387k; the largest decline is the December 2011 drop, which is the truncation artefact rather than a real fall.

Geographic concentration

The United Kingdom contributes £15.99M of £18.86M in total revenue — 84.8% — from 5,410 of the 5,942 identified customers. The export markets behind it are small in customer count but high in value per account.

Table 2 — Largest markets by revenue

COUNTRY	REVENUE	CUSTOMERS	REVENUE PER CUSTOMER
United Kingdom	£15,986,060	5,410	£2,955
EIRE	£609,954	5	£121,991
Netherlands	£548,331	23	£23,841
Germany	£411,959	107	£3,850
France	£321,733	95	£3,387
Australia	£166,444	15	£11,096
Switzerland	£99,403	22	£4,518
Spain	£91,013	41	£2,220

EIRE and the Netherlands stand out sharply. Five identified EIRE accounts generate £610k, and 23 Dutch accounts generate £548k. These are wholesale-scale relationships rather than retail customers, and they behave differently enough that averaging them into the customer base would distort any model that included them uncritically.

Product concentration

Revenue also concentrates on a short head of products. The top four lines alone account for over £1M.

Table 3 — Highest-revenue products			
PRODUCT	REVENUE	UNITS	CUSTOMERS REACHED
Regency Cakestand 3 Tier	£314,045	24,696	1,316
Dotcom Postage	£309,844	2,917	1
White Hanging Heart T-Light Holder	£248,338	94,123	1,494
Jumbo Bag Red White Spotty	£178,401	94,151	978
Party Bunting	£147,080	27,113	894

Two entries are not products at all. *Dotcom Postage* and *Postage* are shipping charges booked as line items, and Dotcom Postage reaches exactly one customer. Any product-level analysis presented commercially should strip these out; they are retained in the modelling tables because removing them would break the revenue reconciliation against the raw extract.

5. Customer segmentation

Segmentation runs on Recency, Frequency and Monetary features derived from the full transaction history. Each is log-scaled with a sign-preserving transform — necessary because Monetary runs negative for customers whose returns exceed their purchases — then standardised before clustering.

K was selected empirically rather than assumed. Silhouette scores were computed for k from 2 to 6:

Table 4 — Cluster count selection by silhouette score	
K	SILHOUETTE SCORE
2	0.4213
3	0.4435
4	0.3663
5	0.3828
6	0.3599

Three clusters scores highest, and the resulting groups are behaviourally distinct rather than cosmetically separated.

Table 5 — Segment profiles				
SEGMENT	CUSTOMERS	AVG REGENCY	AVG FREQUENCY	AVG SPEND

Active core	2,316	42 days	15.5 orders	£6,094
Lapsing / occasional	3,521	300 days	2.5 orders	£645
Net-negative	105	512 days	2.1 orders	−£882

The **active core** is 39% of the customer base and carries almost all repeat revenue. The **lapsing** group is the majority — 59% — but averages two and a half orders and has been silent for roughly ten months; this is the reactivation pool, and its size means even a small conversion rate moves meaningful revenue. The **net-negative** group is small but worth attention: 105 customers whose returns outweigh their purchases. Re-targeting them without first understanding why would actively lose money.

6. Repeat-purchase prediction

Target design

The most consequential methodological decision in the project is how the prediction target is defined. A naive approach would sample customers at random and label them by whether they ever purchased again — which lets future behaviour leak into the features and inflates every metric downstream.

Instead, the target is defined across a time boundary. A cut-off is set 90 days before the end of the data. Features are computed *only* from transactions on or before that cut-off. The label is whether the customer placed an order in the 90 days that followed.

WHY THIS MATTERS

Every reported metric in this section is therefore an estimate of genuine forward-looking performance. The cost is a smaller usable population — 5,346 customers rather than 5,942 — and a harder problem. That trade is worth making: a model validated on leaked data looks excellent and fails in production.

Of the 5,346 customers in the modelling set, 2,322 (43.4%) repurchased within the window. The two groups differ sharply on every behavioural dimension measured before the cut-off:

Table 6 — Pre-cutoff behaviour by outcome

MEASURE	DID NOT RETURN	RETURNED
Customers	3,024	2,322
Average historic spend	£963.61	£4,400.40
Average orders	3.60	11.29
Average distinct products	44.4	111.8

Model benchmark

Six candidates were evaluated under an identical protocol: five-fold stratified cross-validation

on the training split, with PR-AUC as the selection metric. PR-AUC is the right choice for a partially imbalanced target where the positive class is the one of interest; ROC-AUC can look flattering when the negative class dominates.

Table 7 — Five-fold cross-validated performance, ranked by PR-AUC			
MODEL	ROC-AUC	PR-AUC	F1
Gradient Boosting	0.8059	0.7703	0.6750
MLP (64, 32)	0.8041	0.7676	0.6821
MLP (64, 32, 16)	0.8043	0.7667	0.6753
Logistic Regression	0.8024	0.7635	0.6990
Random Forest	0.7904	0.7575	0.6690
MLP (32, 16)	0.8004	0.7571	0.6872

The spread is remarkably tight: every candidate lands within 0.013 PR-AUC of the best. That is itself a finding. When six structurally different algorithms — a linear model, two tree ensembles and three neural networks — agree this closely, the signal is in the features rather than the model family, and further algorithmic search has low expected return.

Notably, the multi-layer perceptrons do not win. This is consistent with the general pattern that neural networks need either substantially more data or richer inputs — sequential, embedded, high-cardinality — to outperform a well-tuned gradient boosting model on eight tabular features.

Final evaluation

Gradient Boosting was tuned by grid search on average precision. The selected configuration is deliberately shallow: 100 estimators, learning rate 0.05, maximum depth 3. It was then evaluated once on the held-out test set, which had been untouched during selection and tuning.

0.800	0.772	0.714	0.622	0.664
ROC-AUC	PR-AUC	Precision	Recall	F1

The test PR-AUC of 0.772 sits almost exactly on the cross-validated 0.770. That agreement is the reassurance worth taking from this section: there is no optimistic bias introduced by model selection and no sign of overfitting to the training folds.

Precision and recall are threshold-dependent and should be read as operational settings rather than fixed properties. At the default 0.5 threshold, roughly seven in ten flagged customers do return, and the model catches 62% of those who would. The threshold is a business dial: raise it when the intervention is expensive and a wasted contact costs real money, lower it when outreach is cheap and missing a churner is the worse error.

A direct comparison against the best MLP confirms the selection held on unseen data:

Table 8 — Held-out test set: champion versus best neural alternative

MODEL	ROC-AUC	PR-AUC	PRECISION	RECALL	F1
Gradient Boosting	0.800	0.772	0.714	0.622	0.664
MLP (64, 32)	0.796	0.764	0.733	0.609	0.665

7. What drives repeat purchase

Two independent methods were used to attribute the model's behaviour, and they agree on the ordering — which matters more than either result alone.

Table 9 — Feature attribution by two methods		
FEATURE	MODEL IMPORTANCE	PERMUTATION IMPORTANCE
Days since last order (recency)	0.599	0.171
Frequency	0.135	0.060
Total revenue	0.109	0.034
Unique products	0.060	0.019
Lifetime days	0.047	0.014
Total items	0.037	0.007
Average transaction value	0.010	0.003
Return count	0.003	0.002

Model importance is computed from the fitted trees; permutation importance measures the drop in test-set PR-AUC when each feature is randomly shuffled, making it model-agnostic and computed on data the model never trained on. Recency dominates both.

The practical consequence is that **silence is the earliest and cheapest churn signal available**. It requires no new data collection, no survey instrument and no behavioural tracking beyond what any order system already records. Intervention timing can be driven straight off the last-order date, with frequency and historic spend used to size the offer rather than to decide whether to make one.

The weakness of return count as a predictor is also worth noting. Returns are a poor signal of impending churn in this dataset — customers who return items are not meaningfully more likely to leave. That runs against a common merchandising assumption and suggests returns policy should be evaluated on margin rather than on retention risk.

8. Recommendation systems

Evaluation protocol

An implicit-feedback interaction table was built from the transaction log: one row per customer-product pair, dated by first adoption, excluding returns, cancellations and non-positive

quantities. Customers and products with fewer than five interactions were filtered out iteratively — filtering one side thins the other, so the pass is repeated.

Evaluation uses a **leave-last-out** protocol. Each customer's most recently adopted product is held back as the target; everything earlier is training data. A recommender scores all products, already-purchased items are masked to negative infinity so no model can win by recommending something the customer already owns, and the top ten are checked against the held-out target.

5,544 Customers	4,206 Products	474,684 Training interactions	5,544 Held-out targets
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Results

Four recommenders were compared on identical targets and metrics: a non-personalised popularity baseline, classical latent-factor matrix factorisation via truncated SVD, an embedding-based Neural Collaborative Filtering model, and a recurrent sequence model for next-item prediction. Both neural models were implemented from scratch in NumPy — forward pass, backpropagation and the Adam optimiser written explicitly — because no deep-learning framework was available in the environment.

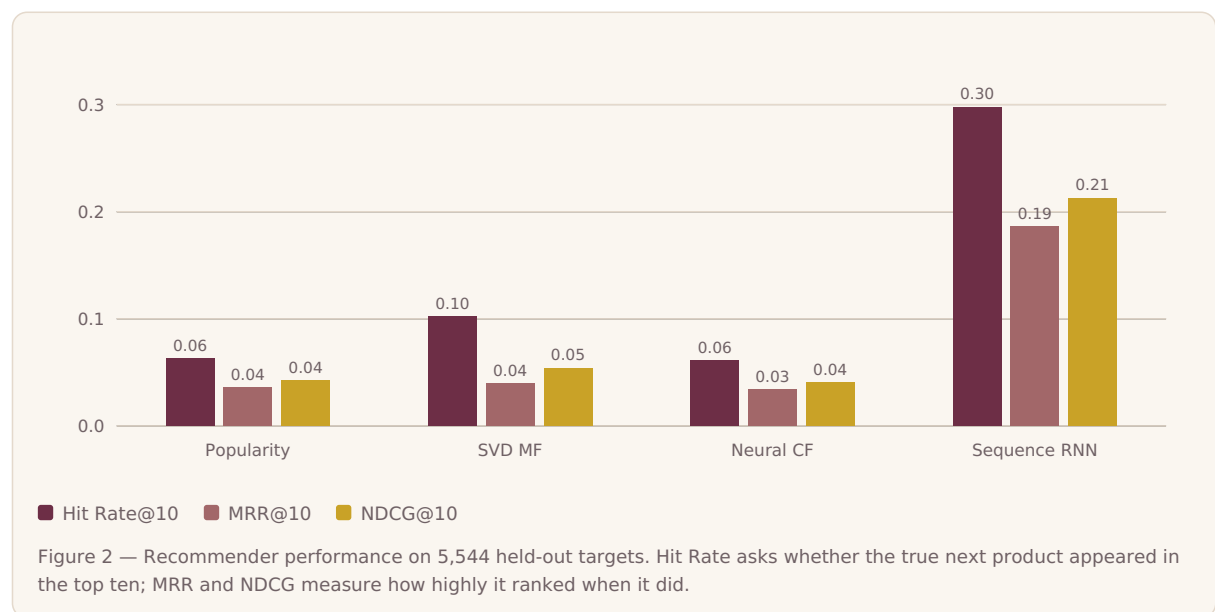


Table 10 — Recommender comparison, leave-last-out

MODEL	TYPE	HITRATE@10	MRR@10	NDCG@10
Popularity	Non-personalised baseline	0.0633	0.0360	0.0423
SVD Matrix Factorisation	Classical collaborative filtering	0.1019	0.0397	0.0541
Neural CF	Neural recommender	0.0613	0.0339	0.0403
Sequence RNN	Neural, sequential	0.2978	0.1863	0.2128

Interpretation

The sequence model wins decisively — $2.9\times$ the classical baseline on Hit Rate, and a larger margin still on MRR, meaning it does not merely surface the right product but ranks it highly. Equally informative: **Neural Collaborative Filtering underperformed the classical SVD baseline**, landing roughly level with the naive popularity model.

That pairing is what makes the result interpretable. If the gap were simply "neural beats classical", both neural models would have won. They did not. The difference between the winner and the rest is that the sequence model has access to *order* — what the customer bought most recently — which the collaborative-filtering models discard when they collapse a purchase history into an unordered set.

DO NOT OVER-READ THIS

The margin reflects strong short-term repeat-purchase behaviour in this specific catalogue: customers restock consumable gift lines in predictable runs, and recency of adoption is highly informative about what comes next. It is not a general claim that recurrent networks beat matrix factorisation. Both neural models were kept deliberately small — few epochs, compact embeddings, CPU-bound — so these figures are floors rather than ceilings, and the ranking between them could shift with proper scaling.

9. Market basket analysis

An association-rule analysis complements the recommenders by asking a different question: not what a customer will buy next, but which products travel together.

The implementation is direct rather than library-based. Customer baskets are built as sets of distinct products, restricted to customers holding at least two. Products clearing a 1% support floor are retained — 2,224 of them — and all pairs within each basket are counted. Support, confidence and lift are computed for both directions of each pair, and rules clearing 10% confidence and lift above 1 are kept. This produced 122,750 rules from 481,947 valid customer-product interactions across 5,736 baskets.

Table 11 — Strongest rules by lift

RULE	SUPPORT	CONFIDENCE	LIFT
French Blue Metal Door Sign 7 → Sign 3	1.24%	84.5%	52.7×
French Blue Metal Door Sign 7 → Sign 8	1.08%	73.8%	48.1×
French Blue Metal Door Sign 0 → Sign 7	0.94%	65.9%	45.0×
French Blue Metal Door Sign 8 → Sign 3	1.10%	71.6%	44.6×
Party Pizza Dish blue+white → pink+white	0.54%	50.0%	44.1×

Two caveats that change the reading

The baskets are lifetime, not per-cart. Because a basket is a customer's full set of distinct products rather than a single invoice, these rules measure affinity *over a customer's history*, not same-basket co-purchase. That is a legitimate quantity — it supports lifecycle merchandising

and follow-up campaigns — but it is not the "customers who bought X also bought Y in the same order" statistic the phrase "market basket" usually implies. Grouping by `invoice_no` instead of `customer_id` would produce the stricter version, and the rules would change substantially.

Lift is inflated by low support. Values above $40\times$ look extraordinary, but these are niche variants sitting just above the 1% support floor, where lift rises naturally because the denominator is small. Confidence is the more operationally meaningful number here: an 84.5% confidence means that of customers who bought Door Sign 7, 84.5% also bought Door Sign 3 at some point. That figure holds up under scrutiny in a way that " $52.7\times$ lift" does not.

The actionable reading

The top rules pair colour and number variants within a single product family, which is a predictable artefact of variant-level SKUs. Rather than being a disappointment, it points at a specific behaviour: **customers work through a set**. Someone who buys one door sign reliably returns for the others; someone who buys the blue pizza dish buys the pink one. That makes bundle offers and "complete the set" prompts the obvious merchandising move, and it gives the retention campaigns a concrete piece of creative to use rather than a generic discount.

10. Engineering and production readiness

Several components exist to make the analysis operable rather than merely correct once.

Automated tests. Four unit tests run inline and pass: mandatory columns contain no nulls and absolute revenue is non-negative; customer order counts and lifetimes are logically coherent with no duplicate customer IDs; the temporal cut-off boundary is strictly enforced in both directions, so no historical feature contains a post-cutoff date and no future target set contains a pre-cutoff one; and model output probabilities are bounded in $[0, 1]$ with correct output length. The third of these is the most valuable — it is an executable assertion that the leakage protection described in section 6 actually holds.

Explainability. Permutation importance runs on held-out data with ten repeats, and SHAP hooks are in place for per-customer attribution where the library is available.

Logging and monitoring. Structured pipeline logging writes to a persistent log file. A monitoring snapshot captures prediction volume, positive prediction rate (37.9%), mean predicted probability (0.4298), standard deviation (0.2607) and the full probability range (0.055 to 0.942). These are the baseline values against which future drift would be measured.

Reproducibility. Random state is fixed at 42 throughout. Preprocessing is fitted on training data only. The complete fitted pipeline — imputation, scaling and estimator together — is persisted with joblib and versioned alongside a metadata file recording model name, version, task, timestamp, metrics and framework.

Scale path. A Spark aggregation path sits alongside the pandas pipeline, guarded so it degrades cleanly when PySpark is absent. It demonstrates the migration route for when the transaction log outgrows a single machine rather than committing the whole pipeline to distributed processing prematurely.

11. Findings and recommendations

Defend the active core

2,316 customers carry the repeat revenue, averaging £6,094 against £645 for the lapsing majority. Recency-triggered contact is the highest-leverage lever available, and the model provides the trigger. Start here rather than with acquisition: the base is already segmented and the intervention is cheap.

Reactivate on silence, not on the calendar

Days since last order drives roughly 60% of the churn prediction. Campaign timing should be driven off each customer's own clock rather than a monthly send schedule. The lapsing segment's average recency of 300 days suggests the current intervention, whatever it is, fires far too late.

Ship the sequence recommender first

29.8% Hit Rate@10 against 10.2% for the classical baseline is the clearest performance gap in the project, and it was achieved with a deliberately undersized model. Scaling it properly — more epochs, larger embeddings, GPU training — is the highest-expected-value engineering task identified here.

Bundle the sets

Variant families show confidence up to 84.5%. Buying one variant strongly predicts returning for the rest, which makes set-completion bundles and prompts a concrete, low-risk merchandising action that requires no new modelling infrastructure.

Reduce concentration risk

84.8% of revenue is UK-only. EIRE and the Netherlands already convert at extraordinary value per account — £122k and £24k respectively — on tiny customer counts. Whether that reflects a replicable wholesale motion or a handful of irreplaceable relationships is the commercial question worth answering next, and it is answerable from the existing data.

12. Limitations

- **Untracked transactions.** 22.8% of transactions carry no customer ID and sit outside every customer-level model. If guest checkouts behave differently from identified customers — which is plausible — the segmentation and churn findings describe only the tracked population.
- **Truncated final month.** The source data ends on 9 December 2011. December 2011 revenue is partial and must not be read as a decline.
- **Undersized neural models.** Both the NCF and RNN were trained for a handful of epochs with compact embeddings on CPU. Their results are lower bounds, and the relative ranking between them is not settled.
- **Basket definition.** As discussed in section 9, association rules are computed over lifetime product sets rather than individual invoices.
- **Single retailer, single period.** Findings describe one UK gift retailer across 2009–2011. Seasonality, category mix and the recency dominance observed here should be re-validated before being transferred to another business.
- **No price or promotion features.** The churn model sees behaviour but not the commercial

context that shaped it — discounting, campaign exposure and stock availability are all absent, and any of them could be confounding the recency signal.

13. Appendix: reproducibility

The full analysis runs end to end from the raw archive. Key parameters:

Table 12 — Analysis parameters		
STAGE	PARAMETER	VALUE
Global	Random state	42
Churn target	Prediction window	90 days after cut-off
Churn model	Train / test split	80 / 20, stratified
Churn model	Cross-validation	5-fold stratified, selection on PR-AUC
Champion	Hyperparameters	100 estimators, lr 0.05, max depth 3
Segmentation	Features	Log-scaled RFM, standardised
Segmentation	k selection	Silhouette, $k \in [2, 6]$
Recommenders	Interaction floor	≥ 5 per customer and per product
Recommenders	Evaluation	Leave-last-out, top-10
Association rules	Thresholds	Support $\geq 1\%$, confidence $\geq 10\%$, lift > 1

Artefacts written by the pipeline include the four processed tables, a data quality report, a generated data dictionary, RFM and segment assignments, cross-validation and final metrics for every model, the recommender comparison, the association rules, the serialised champion pipeline, and a versioned model metadata file.